ABC Church

System Design Document

Version 1.0

04/28/2026

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1. Introduction

The ABC Church Information System (ACCIS) is designed to replace manual paper-based processes with a computerized system to manage membership, contributions, pledges, and ceremonies. ABC Church has experienced rapid growth (400+ households), making manual recordkeeping inefficient and error prone.

1.1 Purpose of the SDD

The purpose of this SDD is to define the system architecture, database design, interfaces, and operational processes required to build the ABC Church Information System.

It serves as guidance for developers, project managers, and stakeholders to ensure the system meets functional and reporting needs.

2. General Overview and Design Guidelines/Approach

This section describes the principles and strategies to be used as guidelines when designing and implementing the system.

2.1 General Overview

ABC Church currently relies on manual, paper-based processes to manage member records, track financial contributions, and document ceremonies. The church has grown significantly to over 400 households, which has made these manual processes inefficient and difficult to maintain accurately. The proposed ABC Church Information System (ACCIS) will replace these manual methods with a centralized computerized system that integrates member management, contribution tracking, pledge management, and ceremony recordkeeping. The system will also support reporting requirements, including annual tax summaries for members and monthly financial and activity reports for the regional headquarters in Detroit.

2.2 Assumptions/Constraints/Risks

2.2.1 Assumptions

It is assumed that the administrative staff and church leadership will receive basic training and be able to effectively use the system. The system is also designed under the assumption that most financial contributions are made at the household level rather than individually, and that each household will maintain a single pledge for the capital campaign. Additionally, it is assumed that the church will continue to operate with a limited number of computers and users at any given time.

2.2.2 Constraints

The design of the system is constrained by the church's limited financial resources, as it operates as a non-profit organization. Staffing constraints also exist, as the church employs only a part-time administrative assistant and a business manager to handle clerical and financial responsibilities. Hardware limitations are also present, as the church currently operates with minimal computer equipment, requiring the system to function efficiently on standard desktop machines without the need for advanced infrastructure.

- Hardware or software environment
- End-user environment
- Availability or volatility of resources
- Standards compliance
- Interoperability requirements
- Interface/protocol requirements
- Licensing requirements
- Data repository and distribution requirements
- Security requirements (or other such regulations)
- Memory or other capacity limitations
- Performance requirements
- Network communications
- Verification and validation requirements (testing)
- Other means of addressing quality goals

2.2.3 Risks

One major risk associated with the system is the potential for data entry errors during the transition from paper records to a digital system. Another risk involves resistance to change from staff who are accustomed to manual processes. Additionally, there is a risk of system downtime or data loss if proper backup and recovery procedures are not implemented. These risks can be mitigated through training, validation controls, and regular data backups.

2.3 Alignment with Federal Enterprise Architecture

The ABC Church Information System does not require alignment with Federal Enterprise Architecture standards, as it is designed for a private, non-profit religious organization and operates independently of federal systems or regulations.

3. Design Considerations

3.1 Goals and Guidelines

The primary goal of the system is to improve the accuracy and efficiency of managing church operations, particularly in tracking financial contributions and maintaining member records. The system is designed to reduce manual workload, streamline administrative processes, and provide timely and accurate reporting. Another key guideline is usability, ensuring that the system interface is simple and intuitive for non-technical users. Data security and integrity are also critical priorities, given the sensitive nature of financial and personal information stored within the system.

3.2 Development Methods & Contingencies

The system will be developed using a structured Systems Development Life Cycle (SDLC) approach, incorporating elements of both the waterfall model and prototyping. This approach allows for clear phases of planning, analysis, design, development, testing, and implementation. Prototyping will be used to refine user interface designs and gather feedback from church staff. In the event of system failure or delays, the church will maintain its existing manual processes as a temporary contingency plan to ensure continuity of operations.

3.3 Architectural Strategies

The system will follow a three-tier architecture consisting of a presentation layer, an application layer, and a data layer.

The presentation layer will provide the user interface through which staff interact with the system. The application layer will handle business logic, such as processing contributions and updating pledge balances. The data layer will consist of a relational database that stores all member, financial, and ceremony data. This architecture was selected because it provides scalability, maintainability, and clear separation of responsibilities within the system.

- Use of a particular type of product (programming language, database, library, commercial off-the-shelf (COTS) product, etc.)
- Reuse of existing software components to implement various parts/features of the system
- Future plans for extending or enhancing the software
- User interface paradigms (or system input and output models)
- Hardware and/or software interface paradigms
- Error detection and recovery
- Memory management policies
- External databases and/or data storage management and persistence
- Distributed data or control over a network
- Generalized approaches to control
- Concurrency and synchronization
- Communication mechanisms
- Management of other resources

3.4 Performance Engineering

The system is designed to handle the operational needs of approximately 400 households, with a relatively low number of concurrent users, typically between one and three administrative users at any given time. Performance requirements are modest, with an emphasis on fast data retrieval and efficient report generation. Monthly and annual reports should be generated within seconds, and data entry operations should be processed in real time without noticeable delays.

Performance requirements are a contributing factor to system design.

Performance requirements are the defined scalability or responsiveness expectations of specific workloads that process on a system.

4. System Architecture and Architecture Design

The ABC Church Information System (ACCIS) is designed using a modular, three-tier architecture that separates the system into presentation, application, and data layers. This structure allows data to flow efficiently between user interfaces, business logic, and the central database while maintaining consistency and scalability. The system interacts with external entities such as church members, administrative staff, financial institutions, and regional headquarters by receiving input data and generating reports. At a high level, the system is divided into key subsystems, including Member Management, Contribution Management, Pledge Management, Ceremony Management, and Reporting. Each subsystem is responsible for a specific function but shares data through a centralized database, ensuring that information entered in one area is accessible across the system. For example, contribution data is used by the reporting subsystem to generate tax summaries and financial reports.

The architecture was chosen for its simplicity, maintainability, and ability to support future growth. A three-tier design was preferred over a monolithic approach to improve organization and flexibility. The system also applies common design patterns such as Model-View-Controller (MVC) and layered architecture to separate responsibilities and streamline data flow between components.

This section outlines the system and hardware architecture design of the system.

4.1 Logical View

The system is logically divided into several functional modules, including member management, contribution tracking, ceremony management, and reporting. Each module is responsible for a specific set of tasks, but all modules share access to a centralized database. The member management module handles household and individual member data, while the contribution tracking module records weekly donations and capital campaign contributions. The ceremony management module maintains records of baptisms, weddings, and funerals, and the reporting module generates financial and administrative reports.

4.2 Hardware Architecture

The system will operate on a small-scale hardware setup appropriate for a non-profit organization. This includes standard desktop computers located in the church office for administrative use, along with a printer for generating reports and documentation. A network router will provide connectivity between devices, and an external storage device will be used for backups. The hardware configuration is designed to be cost-effective while still meeting the operational needs of the system.

The system will use a small, low-cost hardware setup that fits ABC Church's needs as a non-profit organization. The main equipment will be located in the church office and will support daily data entry, record keeping, contribution tracking, and report printing.

- The setup will include:
  - 1 main desktop computer to host the system/database
  - 1–2 staff desktop computers for daily use
  - 1 printer for reports and tax summaries
  - 1 router/switch for network connection
  - 1 external drive or cloud backup option
- Processor capacity should be basic but reliable, such as:
  - Intel i5 or similar processor
  - enough speed for data entry, searches, and report generation
- Memory estimate:
  - 8 GB RAM minimum
  - 16 GB preferred for the main computer
- Online storage estimate:
  - 256 GB minimum internal storage
  - 512 GB preferred for the main computer
- Auxiliary storage estimate:
  - 1 TB external backup drive
  - optional cloud backup for extra protection
- The printer will be used for:
  - contribution summaries
  - monthly reports
  - annual tax reports
  - церемoney and membership records
- This setup is enough because the church only needs to manage around 400 households and a small number of users at one time.

- The hardware design keeps costs low while still supporting reliable daily operations.

4.2.1 Security Hardware Architecture

Security at the hardware level will be supported through the use of a firewall-enabled router to protect the network from external threats. Access to the system will be restricted to authorized users through password-protected computers. Backup devices will be stored securely to prevent unauthorized access or data loss.

4.2.2 Performance Hardware Architecture

The system does not require high-performance hardware due to its relatively low processing demands. Standard office computers with adequate memory and storage capacity will be sufficient to support system operations. This ensures that the system remains cost-effective while maintaining reliable performance.

4.3 Software Architecture

The system will be supported by standard operating systems such as Windows and will utilize a relational database management system such as Microsoft Access or SQL Server. The application software will be custom developed to meet the specific needs of the church. Reporting tools will be integrated into the system to allow for the generation of financial and administrative reports.

4.3.1 Security Software Architecture

Security within the software will be implemented through user authentication and role-based access control. Users will be required to log in with a username and password, and access to system functions will be restricted based on user roles, such as administrator or staff member.

4.3.2 Performance Software Architecture

The software will be designed to optimize performance through efficient database queries and minimal system overhead. Given the relatively small scale of the system, performance optimization will focus on ensuring quick response times for data entry and report generation.

4.4 Information Architecture

The system will store structured data related to households, individual members, financial contributions, pledges, and money records. Much of this data qualifies personally identifiable information (PII), including names, addresses, and financial contributions. As a result, appropriate measures will be taken to ensure data privacy and security.

4.4.1 Records Management

The system will maintain records of financial contributions for tax reporting purposes, as well as historical records of membership and monies. These records will be retained according to organizational needs, with regular backups to ensure data preservation and recovery in case of system failure. Federal regulations issued by the National Archives and Records Administration (NARA) are outlined in 36

Code of Federal Regulations (CFR) - Subchapter B - Records Management. Business owners must contact the Office of Strategic Operations and Regulatory Affairs (OSORA) to initiate the record management process.

4.4.1.1 Data

The system uses a mix of paper, manual entry, and electronic data. Most of the data comes from church members and staff and is entered into the system for storage and reporting.

- Data comes from:
  - church members (registration info, contributions, pledges)
  - administrative staff (data entry and updates)
  - церемoney records (baptisms, weddings, funerals)
- Data formats include:
  - paper forms (member sign-ups, церемoney forms)
  - manual input (typed into the system)
  - electronic data (stored in the database and reports)
- Types of data collected:
  - household and member details
  - contribution amounts and dates
  - pledge commitments and balances
  - церемoney information

4.4.1.2 Manual/Electronic Inputs

After data is entered into the system, it is checked for accuracy and then stored in the main database. This ensures that all records are correct and ready for use across the system.

- Data is entered by staff through input screens based on paper forms or direct input.
- The system checks the data for:
  - missing fields
  - correct formats (dates, amounts, contact info)
- Once verified, the data is:
  - saved into the database
  - made available to all system modules
- Temporary input data is cleared after it is stored.
- If errors are found, users are prompted to fix them before saving.

4.4.1.3 Master Files

The system stores all main data in master files (database tables), which hold the core records used by the system for daily operations and reporting.

- The main master files include:
  - household file
  - member file
  - contribution file

  - pledge file
  - pledge contribution file
  - церемoney files (baptism, wedding, funeral)
  - reverend file
- These files store key information such as:
  - names, addresses, and contact details
  - financial contributions and pledge amounts
  - церемoney dates and participants
- Each file is organized using unique IDs and linked together so data can be shared across the system.
- Master files are updated regularly as new data is entered or changes are made.
- These files support:
  - daily operations
  - reporting
  - long-term record keeping

4.5 Internal Communications Architecture

The system will operate on a local area network (LAN) using standard Ethernet connections. This network will allow communication between the administrative computers and the database system, enabling real-time data entry and retrieval.

4.6 Security Architecture

The overall security architecture will include user authentication, data validation, and regular backups. These measures are designed to protect sensitive financial and personal information while ensuring system integrity.

4.7 Performance

The system is expected to provide fast and reliable performance, with minimal delays in data entry and retrieval. Reports should be generated quickly, and the system should be able to handle daily operations without performance issues.

4.8 System Architecture Diagram

The system architecture consists of a three-tier model, including a user interface layer, an application logic layer, and a database layer. These components work together to process user input, perform necessary computations, and store data efficiently.

5. System Design

5.1 Business Requirements

The ABC Church Information System must support the full range of administrative and financial operations required by the church. The system must allow staff to register and maintain records for households and individual members, including personal and contact information. It must track weekly contributions made through pre-numbered envelopes as well as loose offerings. The system must also support the management of capital campaign pledges, including recording the initial pledge amount, tracking contributions toward the pledge, and calculating remaining balances. In addition, the system must maintain detailed records of monies such as baptisms, weddings, and funerals, including participant and officiating minister information. The system must generate reports for internal use, annual tax reporting for members, and monthly submissions to the regional headquarters.

5.2 Database Design

The database is designed as a relational structure that organizes data into multiple related tables to ensure consistency, reduce redundancy, and support efficient querying. The system maintains relationships between households, members, contributions, pledges, and money records. Each table includes a primary key to uniquely identify records and foreign keys to establish relationships between entities.

The primary data objects in the system include households, registered family members, weekly contributions, capital campaign pledges, pledge contributions, baptisms, weddings, funerals, and reverends. Each household serves as the central financial entity and is linked to multiple family members. Contributions and pledges are associated with households, while money records are associated with individual members and the presiding reverend. These relationships allow the system to accurately track both financial and administrative data across different functional areas.

5.2.1 File and Database Structures

The system uses structured relational tables to store data, with clearly defined relationships between entities. Household records are stored in a master table, which is linked to member records through a one-to-many relationship. Contribution and pledge tables are also linked to households, allowing the system to track financial activity at the household level. Ceremony tables are linked to both members and reverends, ensuring that all money data is properly associated with participants and officiants. The database structure supports efficient querying and reporting while maintaining data integrity.

5.2.1.1 Database Management System Files

The system uses a relational database to store all core data, including households, members, contributions, pledges, and ceremony records. This database acts as the main storage area for the system and supports all modules by allowing data to be added, updated, and retrieved quickly and accurately.

- The main database tables include:
  - households
  - members
  - weekly contributions
  - pledges
  - pledge contributions
  - ceremonies (baptism, wedding, funeral)
  - reverends
- Each table stores related data, such as:
  - household info (name, address, phone)
  - member details (name, DOB, role)
  - contribution amounts and dates
  - pledge totals and balances
  - ceremony details and participants
- Each record has a unique ID (primary key), and tables are connected using shared fields (foreign keys).
- Data is stored in structured rows and columns, making it easy to search, update, and report on.
- The database supports:
  - creating new records
  - updating existing records
  - retrieving data for reports
  - deleting outdated or incorrect entries
- Data is accessed directly when needed, so users can quickly look up records without delays.

- The database size is moderate, since it supports around 400 households and their related records.
- Data is updated regularly:
  - contributions entered weekly
  - pledges updated as payments are made
  - member and ceremony records updated as needed
- The system supports multiple users, but since usage is low, performance remains fast and stable.
- Backup and recovery include:
  - regular database backups (daily or weekly)
  - storage on external drives or cloud
  - ability to restore the database if data is lost

Non-Database Management System Files

The system uses a few simple non-database files mainly for reports, backups, and temporary processing. These files support daily operations by storing output reports, keeping backup copies of data, and handling temporary data during system use.

- The main non-database files include:
  - report files (tax summaries, monthly reports) → output
  - backup files → both (save and restore)
  - temporary files → used during processing
- These files are mainly used by:
  - reporting module (creates reports)

- system/backup process (handles backups and recovery)
- File records include basic data like:
  - household names
  - contribution amounts
  - dates
  - pledge totals
- Records can vary in size depending on the report, so they are variable length.
- Files are accessed directly when needed, allowing quick opening and use.
- File sizes stay small to moderate since the church only manages around 400 households.
- Files are updated on a regular basis:
  - weekly for contributions
  - monthly for reports
  - yearly for tax summaries
- System usage is low, so file updates happen smoothly without performance issues.
- Backup and recovery include:
  - regular backups (daily or weekly)
  - storage on external drive or cloud
  - ability to restore lost data if needed

5.3 Data Conversion

Data conversion will involve transferring existing paper-based records into the new system through manual data entry. Historical records, including member information, contribution histories, and money records, will be entered into the database by administrative staff. To ensure accuracy, data entry will be verified through validation checks and review processes. During the transition period, both manual and digital systems may operate simultaneously until all critical data has been successfully migrated.

5.4 User Machine-Readable Interface

The system will provide a user-friendly interface designed for administrative staff with varying levels of technical expertise. Users will interact with the system through forms and menus that allow them to input, update, and retrieve data efficiently. The interface will be designed to minimize errors and provide clear navigation between system functions.

5.4.1 Inputs

System inputs include member registration data, contribution entries, pledge information, and money records. Data will be entered manually through on-screen forms designed to match existing paper forms, making the transition easier for staff. Input validation rules will ensure that required fields are completed, and that data is entered in the correct format, such as dates, phone numbers, and monetary values.

- Copies of form(s) if the input data are keyed or scanned for data entry from printed forms
- Description of any access restrictions or security considerations
- Each transaction name, code, and definition, if the system is a transaction-based processing system

- Incorporation of the Privacy Act statement into the screen flow, if the system is covered by the Privacy Act
- Description of accessibility provisions to comply with Section 508 of the Rehabilitation Act

5.4.2 Outputs

System outputs include a variety of reports and data displays. These outputs include annual contribution summaries for tax purposes, monthly financial reports for regional headquarters, and internal management reports such as pledge status summaries. Reports will be available in both on-screen and printed formats, and access to sensitive financial reports will be restricted to authorized users.

- Identification of codes and names for reports and data display screens
- Description of report and screen contents (provide a graphical representation of each layout and define all data elements associated with the layout or reference the data dictionary)
- Description of the purpose of the output, including identification of the primary users
- Report distribution requirements, if any (include frequency for periodic reports)
- Description of any access restrictions or security considerations

5.5 User Interface Design

The user interface will follow standard design principles to ensure usability and efficiency. Screens will be organized logically, with clear labels and consistent layouts. Navigation will be menu-driven, allowing users to access different system functions easily. The interface will also include error messages and confirmation prompts to guide users and prevent mistakes.

5.5.1 Section 508 Compliance

The system will incorporate basic accessibility features to ensure usability for all users. These features include clear text, simple layouts, and readable font sizes. While full federal compliance is not required, the system will still aim to follow general accessibility best practices.

6. Operational Scenarios

The system supports several key operational scenarios that reflect the daily activities of the church. In a member registration scenario, administrative staff enter household and member information into the system, which is then stored in the database and used for future interactions. In a contribution tracking scenario, staff record weekly donations and campaign contributions, which are automatically linked to the appropriate household and used to update financial records. In a money management scenario, staff record details of baptisms, weddings, and funerals, including participants and officiating ministers. Reporting scenarios allow users to generate financial summaries and administrative reports for both internal and external use.

7. Detailed Design

The detailed design section provides the specifications needed to build and implement the system. It includes descriptions of hardware, software, security, performance, and communication components.

7.1 Hardware Detailed Design

The hardware design includes standard desktop computers for administrative use, a printer for generating reports, and external storage devices for backups. Equipment will be placed in the church office as indicated in the floor plan to ensure accessibility and security. The hardware configuration is designed to support daily operations without requiring advanced infrastructure.

The system uses basic equipment like office desktop computers, a printer, and an external drive for backups, all set up in the church office. It runs on normal power (standard wall outlets), so no special electrical setup is needed. Since this is a small system, things like signal levels and technical wiring details are handled by standard equipment and don't require special setup. Devices connect using common ports like USB and Ethernet, so everything is easy to plug in and set up. Each computer has enough memory and storage to handle member records, contributions, and reports without slowing down.

The computers use standard processors that are fast enough for data entry, searching, and report generation. The setup includes:

- 1 main computer (runs the system and stores data)
- 1–2 staff computers
- 1 printer
- 1 external backup device

7.2 Software Detailed Design

The software is organized into functional modules, including member management, contribution tracking, money management, and reporting. Each module performs specific tasks while interacting with the central database. The system will be developed using a structured programming approach, with clearly defined functions and data flows. The software will also include validation rules, error handling, and reporting capabilities.

The system is split into simple modules like members, contributions, pledges, ceremonies, and reports, all connected to one main database. Each module has a clear job, like storing member info, tracking donations, or creating reports. These modules handle the main needs of the church, like keeping records, tracking money, and printing reports.

Data is stored in basic tables for things like households, members, contributions, pledges, and ceremonies.
The system keeps things simple with:

- a small number of users
- standard computers
- required formats for things like dates and amounts

Each module has smaller tasks inside it, like entering data, updating it, and viewing it later.
All modules share the same data, so:

- member info is used across the system
- contribution data shows up in reports

The system works by:

- entering data
- checking it
- saving it
- pulling it back when needed

It handles errors like:

- missing info
- wrong formats
- duplicate entries

Users interact through simple screens for:

- entering data
- viewing records
- printing reports

The system handles reporting by creating:

- yearly tax summaries
- monthly reports for headquarters

7.3 Security Detailed Design

Security measures include user authentication, role-based access control, and data encryption where necessary. Users must log in with a valid username and password, and access to system functions is restricted based on user roles. Audit logs will track system activity, and regular backups will protect against data loss.

- Authentication
- Authorization
- Logging and Auditing
- Encryption
- Network ports usage
- Intrusion Detection and Prevention (especially if hosted in a non-CMS data center

The design should be based on the designated system's security level and provide adequate protection against threats and vulnerabilities.

7.4 Performance Detailed Design

The system is designed to meet performance requirements with minimal resource usage. It will support quick data entry and retrieval, as well as efficient report generation. Given the limited number of users and moderate data volume, the system will operate effectively on standard hardware without performance issues.

The system is designed to run smoothly with low resource usage and will work well on standard office computers.

It will handle about 400 households and their records, with only 1–3 users at a time, so the overall data load is moderate.

The system is expected to provide:

- fast data entry
- quick lookups
- reports generated within seconds

Since usage is low, the system will stay responsive even during busy times, like after Sunday services.

The design keeps things simple by using:

- a central database
- basic queries for fast results
- minimal background processing

To keep the system available, it will rely on:

- a reliable main computer
- regular system checks by staff

Backups will be done regularly (daily or weekly) using an external drive or cloud storage to prevent data loss.

If something goes wrong, data can be restored from backups, and important records can also be printed or saved as reports.

A possible weak point (single point of failure) is the main computer that stores the database.

Since the system is small, advanced setups like clustering aren't needed, but risk can be reduced by:

- keeping backup copies
- using a secondary computer if needed

7.5 Internal Communications Detailed Design

The system will operate on a local area network that connects administrative computers to the database. Data will be transmitted securely within the network, allowing real-time updates and access. The communication design ensures that all system components can interact efficiently.

- Client PCs → Router/Switch → Main PC (Database Host) → Printer
- The system runs on a **small local network (LAN)** that connects the office computers to one main system.
- There will be **1 main computer (database host)** and about **1–2 other computers** used by staff.
- Everything connects through a **router/switch**, so all computers can share and access data.
- The system uses **standard network connections**, so no special setup is needed.
- Data being shared includes:

- Member info
- Contributions
- Pledges
- Reports

- The network uses a **star layout**, meaning all devices connect to one central point.
- All equipment is **close together (about 10–30 feet)**, so setup is simple and fast.
- Data moves **back and forth between computers**, allowing:

- Real-time updates
- Easy data entry
- Quick report access

8. System Integrity Controls

The system includes multiple integrity controls to ensure data accuracy and security. Input validation rules prevent incorrect data entry, while audit trails track changes to critical data. Backup and recovery procedures ensure that data can be restored in the event of system failure. Access controls restrict sensitive operations to authorized users only.

- Internal security to restrict access of critical data items to only those access types required by users/operators
- Audit procedures to meet control, reporting, and retention period requirements for operational and management reports
- Application audit trails to dynamically audit retrieval access to designated critical data
- Standard tables to be used or requested for validating data fields
- Verification processes for additions, deletions, or updates of critical data
- Ability to identify all audit information by user identification, network terminal identification, date, time, and data accessed or changed.

9. External Interfaces

The system interfaces with external entities primarily through reporting functions. Monthly financial and money reports are submitted to the regional headquarters in Detroit, and annual contribution summaries are provided to members for tax purposes.

9.1 Interface Architecture

The interface architecture supports the exchange of information between the system and external stakeholders through report generation and data export. These interfaces are simple and do not require real-time integration with external systems.

9.2 Interface Detailed Design

The detailed design of external interfaces focuses on report formatting and data accuracy. Reports must include all required financial and administrative data and be generated in formats suitable for submission and distribution. Access to these reports is controlled to ensure confidentiality.

Appendix A: Record of Changes

Use the table below to provide the version number, the date of the version, the author/owner of the version, and a brief description of the reason for creating the revised version.

Table 1 - Record of Changes

Version Number	Date	Author/Owner	Description of Change
1.0	*04/28/2026*	*TMS*	*Initial SDD*

Appendix B: Acronyms

Table 2 - Acronyms

Acronym	Literal Translation
ACCIS	ABC Church Information System
ABC	ABC Church
SDD	System Design Document
SDLC	Systems Development Life Cycle
ERD	Entity-Relationship Diagram
DFD	Data Flow Diagram
DBMS	Database Management System
UI	User Interface

Acronym	Literal Translation
LAN	Local Area Network
HQ	Headquarters
PII	Personally Identifiable Information
CRUD	Create, Read, Update, Delete
PK	Primary Key
FK	Foreign Key
WBS	Work Breakdown Structure
Gantt	Gantt Chart
COTS	Commercial Off-The-Shelf
OS	Operating System

Appendix C: Glossary

Table 3 - Glossary

Term	Acronym	Definition
Application Layer	N/A	The part of the system responsible for processing business logic, such as calculating contributions and updating pledge balances.
Baptism Record	N/A	A stored record of a baptism ceremony, including participant details, godparents, and officiating minister.
Capital Campaign	N/A	A fundraising initiative by the church to raise money for large projects, such as purchasing property or building improvements.
Contribution	N/A	A financial donation made by a household or individual to the church, including weekly offerings and campaign payments.
Context Diagram	DFD Level 0	A high-level diagram that shows the system and its interactions with external entities such as members and headquarters.
Data Flow Diagram	DFD	A diagram that illustrates how data moves through the system,

		including inputs, processes, and outputs.
Database	DB	A structured collection of data stored electronically for easy access, management, and updating.
Entity	N/A	A real-world object or concept represented in the database, such as a household, member, or contribution.
Entity-Relationship Diagram	ERD	A visual representation of the relationships between entities in a database system.
External Interface	N/A	A point of interaction between the system and external entities, such as reporting to regional headquarters.
Household	N/A	A group of individuals registered together as a single unit within the church, typically used for financial tracking.
Information System	IS	A combination of hardware, software, data, and processes used to collect, manage, and process information.
Member	N/A	An individual who is registered as part of a household within the church.

Operational Scenario	N/A	A step-by-step description of how users interact with the system to perform specific tasks.
Pledge	N/A	A financial commitment made by a household to contribute a specific amount to the capital campaign over time.
Pledge Contribution	N/A	A payment made toward fulfilling a previously established pledge.
Primary Key	PK	A unique identifier for a record in a database table.
Foreign Key	FK	A field in a database table that links to the primary key of another table to establish relationships.
Presentation Layer	N/A	The part of the system that provides the user interface allows users to interact with the system.
Relational Database	N/A	A database that organizes data into tables with relationships between them.
Reporting Module	N/A	A system component responsible for generating reports such as tax summaries and financial statements.

Reverend	N/A	A minister who performs monies and is associated with baptism, wedding, and funeral records.
System Architecture	N/A	The overall structure of the system, including hardware, software, and data components.
User Interface	UI	The visual layout and interactive elements through which users interact with the system.
Weekly Contribution	N/A	A regular donation made by a household during church services, typically using pre-numbered envelopes.
Wedding Record	N/A	A stored record of a wedding ceremony, including participant details and officiating minister.
Funeral Record	N/A	A stored record of a funeral ceremony, including deceased member details and information.

Appendix D: Referenced Documents

Table 4 - Referenced Documents

Document Name	Document Location and/or URL	Issuance Date
ABC Church System Analysis and Design Case	Provided course document	N/A
System Design Document (SDD) Template	Provided course template	N/A
Church Floor Plan	Provided project file	N/A
Database Systems: Design, Implementation, and Management (Coronel & Morris)	Course textbook	2018
Essentials of Systems Analysis and Design (Valacich, George, Hoffer)	Course textbook	2014
Microsoft Access Documentation	https://learn.microsoft.com	Current
Lucidchart Diagramming Tool	https://www.lucidchart.com	Current

Appendix E: Approvals

The undersigned acknowledge that they have reviewed the SDD and agree with the information presented within this document. Changes to this SDD will be coordinated with, and approved by, the undersigned, or their designated representatives.

Table 5 – Approvals

Document Approved By	Date Approved
Name: ______________________ Title: Business Manager (Margie Robbens) Organization: ABC Church	______________________
Name: ______________________ Title: Senior Minister (Reverend Timothy Beck) Organization: ABC Church	______________________
Name: ______________________ Title: Project Manager Organization: Information Systems Consulting Team	______________________
Name: ______________________ Title: Systems Analyst (Student) Organization: Project Team	______________________

Appendix F: Additional Appendices

This appendix provides supporting materials that enhance the understanding of the ABC Church Information System design, including diagrams, planning artifacts, and data structures.

F.1 Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram illustrates the relationships between households, members, contributions, pledges, money records, and reverends. The design ensures that all financial and money data are properly linked and normalized to reduce redundancy.

*

Description:
The ERD shows that the household serves as the central entity for financial transactions. Each household may have multiple members and contributions, while money records are associated with individual members and the officiating reverend.

F.2 Context Diagram (Level 0 DFD)

The context diagram provides a high-level view of the system and its interaction with external entities.

*

Description:
The system interacts with church members (for contributions and registration), administrative staff (for data entry and reporting), the bank (for loan and financial tracking), and the regional headquarters (for monthly reporting).

F.3 Data Flow Diagram (Level 1 DFD)

The Level 1 DFD breaks down the system into its main processes and shows how data flows between them.

*

Description:
The diagram illustrates processes such as member registration, contribution processing, pledge tracking, and церemoney management. Data flows between these processes and the central database.

F.4 Work Breakdown Structure (WBS)

The Work Breakdown Structure defines the hierarchical breakdown of project tasks.

*

Summary Description:
The WBS includes project phases such as planning, analysis, design, development, testing, deployment, and

maintenance. Each phase is divided into smaller tasks to ensure clear project organization and execution.

*

F.5 Gantt Chart

The Gantt Chart provides a timeline of project activities, showing task durations and sequencing.

*

Summary Description:
The chart outlines the project schedule, including major phases such as system analysis, design, development, testing, and implementation. It helps track progress and ensure timely completion.